Weekly Progress Report

Week 1 (January 10 – January 16, 2026) — **University Agent & Sitemaps**

During the first week of the internship, the primary focus was on establishing the foundational architecture for the university agent service and integrating it into the backend infrastructure. The Video Interview v2 module was initialized with a new agent architecture, providing the structural basis for subsequent AI-driven interview features. The university agent was developed with robust abbreviation handling and simplified selection logic, alongside comprehensive edge case handling to ensure reliable university matching. Environment variable configuration was adopted for the agent URL, promoting configurability across deployment environments. Additionally, dynamic sitemaps and a robots.txt file were implemented to strengthen the platform's search engine optimization (SEO) posture.

Date	Task
Jan 10	Initialized Video Interview v2 module with new agent architecture
Jan 12	Built university agent with abbreviation handling and simplified university selection logic
Jan 13	Added edge case handling and extended abbreviation support for university matching
Jan 14	Moved university agent URL to environment variables for configurability
Jan 15	Fixed TypeScript error in university controller
Jan 16	Added dynamic sitemaps and robots.txt with sitemap configuration

> **Week Summary:** *During the first week of the internship, the primary focus was on establishing the foundational architecture for the university agent service and integrating it into the backend infrastructure.*

Week 2 (January 17 – January 23, 2026) — **Video Interview v2 & AI Voice Integration**

The second week was dedicated to completing the core Video Interview v2 system, which represented a significant architectural advancement over the previous iteration. A new interview agent was constructed with an updated architecture, supplemented by detailed logging throughout the interview flow to facilitate debugging and monitoring. Database migrations were created and interfaces updated to support the new interview data model. Critical bugs, including a duplicate question issue and report generation failure, were identified and resolved. The time limit for video interviews was extended to twenty minutes to accommodate comprehensive interview sessions. The integration of AI voice through Text-to-Speech technology into the interview agent marked a significant milestone, enabling a conversational interview experience. The Google Gemini API was also integrated for dynamic question generation, replacing the previous static question approach, while the user profile was enhanced with a dedicated Work Experience section.

Date	Task
Jan 18	Built new interview agent with updated architecture
Jan 19	Added detailed logging to video interview flow
Jan 19	Created database migration and updated interfaces for interview v2
Jan 20	Fixed duplicate question bug and report generation issue in video interview
Jan 20	Fixed video access permissions and corrected typographical errors
Jan 20	Increased video interview time limit to 20 minutes
Jan 21	Redesigned interview report UI to be more comprehensive and detailed
Jan 22	Integrated AI voice (Text-to-Speech) into the interview agent
Jan 23	Fixed About Me page and updated to new Gemini API
Jan 23	Added Work Experience section to the About Me profile page

> **Week Summary:** *The second week was dedicated to completing the core Video Interview v2 system, which represented a significant architectural advancement over the previous iteration.*

Week 3 (January 24 – January 30, 2026) — **Dynamic Interview & Admin Panel Improvements**

The third week focused on stabilising the dynamic AI interview system and improving the administrative panel's functionality. User interface issues in the AI voice agent interface were resolved, and the dynamic interview process was refined with an increased follow-up question token size to support more detailed responses. A critical race condition in the video interview audio integration was identified and corrected, ensuring consistent audio playback during interview sessions. A new game-style report was integrated into the interview flow to enhance user engagement with their results. The admin panel received several improvements, including a new reports section and updated onboarding questions. Security and DevOps documentation was authored, establishing formal operational guidelines for the platform.

Date	Task
Jan 24	Fixed UI issues for the AI voice agent interface
Jan 26	Updated dynamic interview process and increased follow-up question token size
Jan 26	Fixed parsing of follow-up questions from AI response
Jan 26	Integrated new game-style report into the interview flow
Jan 26	Added new reports section to admin panel
Jan 27	Fixed admin panel issues and production deployment bugs
Jan 27	Resolved race condition in video interview audio integration
Jan 28	Implemented new admin panel changes including new onboarding questions
Jan 28	Fixed country of origin field issue in mentor profile
Jan 29	Fixed voice agent and WebSocket connection stability
Jan 30	Added security and DevOps documentation

> **Week Summary:** *The third week focused on stabilising the dynamic AI interview system and improving the administrative panel's functionality.*

Week 4 (January 31 – February 6, 2026) — **Security Hardening & Real-time Transcription**

Security hardening was the primary objective of the fourth week, encompassing both administrative access controls and user authentication mechanisms. The admin panel was secured with login attempt tracking tables to detect and prevent brute force attacks, while a user authentication security layer was implemented across the platform. Phone number validation improvements were deployed to ensure data integrity during the mentor signup process. A major technical achievement was the integration of Google Cloud's Chirp 3 speech-to-text API, enabling real-time transcription of interview audio via a dedicated Cloud Run service. IP-based geolocation was added to support personalised user experiences, and the AI interview prompt was refined to produce more contextually appropriate follow-up questions.

Date	Task
Jan 31	Added admin panel security with login attempt tracking tables
Jan 31	Implemented user authentication security layer
Feb 2	Fixed phone number validation message display
Feb 2	Completed UI improvements across mentor onboarding
Feb 3	Integrated real-time Chirp 3 speech-to-text transcription via Cloud API
Feb 3	Added background transcription Cloud Run API
Feb 3	Fixed phone validation issue in mentor signup
Feb 3	Added auto-scrolling behaviour for mentoring service error section
Feb 3	Removed unused files and cleaned up console warnings
Feb 4	Added IP-based geolocation and fixed database column
Feb 6	Updated AI interview prompt for better intent follow-up

> **Week Summary:** *Security hardening was the primary objective of the fourth week, encompassing both administrative access controls and user authentication mechanisms.*

Week 5 (February 7 – February 13, 2026) — **Security v2, Sentry & Background Video Upload**

The fifth week completed the second phase of security enhancements and introduced professional-grade error monitoring. Cookie-based authentication was implemented to replace less secure session handling mechanisms, accompanied by rate limiting and same-origin policy enforcement for Cloud Run services. API response sanitisation was applied to remove sensitive user identifiers from public-facing responses. Sentry was integrated across both the frontend and backend to provide real-time error tracking and alerting in production. On the payment side, a Razorpay webhook handler was developed alongside the necessary database migrations, establishing the infrastructure for future payment processing. Multiple user interface bugs in the mentor dashboard were also resolved during this period.

Date	Task
Feb 4	Implemented video auto-stop recording after 2 minutes
Feb 4	Fixed admin dashboard UI — duration display with fallback for 0s videos
Feb 4	Added duration field insertion during video upload
Feb 5	Added dynamic interview controller to backend
Feb 5	Fixed TypeScript errors in dynamic interview component
Feb 8	Added cookie-based authentication and fixed sign-in/sign-up flow
Feb 8	Introduced rate limiting, same-origin policy for Cloud Run services
Feb 8	Added response sanitisation and removed userId from API responses
Feb 8	Integrated Sentry error tracking for frontend and backend
Feb 8	Applied production deployment fixes
Feb 9	Fixed Work Experience UI and resolved hourly rate = 0 bug
Feb 9	Made report character card content scrollable
Feb 9	Fixed approve/dashboard button UI and admin panel pane toggle
Feb 9	Added inactive section state after mentor report generation
Feb 10	Added deployment script for production builds
Feb 12	Resolved minor Sentry error in production
Feb 13	Added Razorpay webhook handler and database migrations

> **Week Summary:** *The fifth week completed the second phase of security enhancements and introduced professional-grade error monitoring.*

Week 6 (February 14 – February 20, 2026) — **Email Funnel System**

The sixth week was entirely devoted to designing and building a comprehensive email funnel system from the ground up. The system encompassed campaign creation, scheduling, and delivery logic, supported by a cron job mechanism for automated and time-triggered email dispatch. Administrative controls were added to the admin panel to facilitate template management, campaign toggling, and monitoring. Razorpay webhook signature verification was implemented to ensure the integrity of payment event notifications. Several critical bugs were addressed, including a foreign key constraint violation in the email messages table and session storage issues affecting mentor Google OAuth sign-ups. Student onboarding flow defects were also corrected in both development and production environments.

Date	Task
Feb 14	Built email funnel system — campaign creation, scheduling, and delivery
Feb 16	Added email templates management in admin panel
Feb 16	Optimised email funnel delivery logic
Feb 17	Implemented Razorpay webhook signature verification
Feb 17	Added email funnel environment configuration
Feb 18	Fixed mentor Google OAuth signup session storage issue
Feb 18	Fixed student onboarding flow
Feb 18	Updated Google Search Console verification file
Feb 19	Fixed foreign key constraint issue with email messages table
Feb 19	Reopened email cron job for scheduled sending
Feb 19	Added admin panel email toggle switch (enable/disable campaigns)
Feb 19	Added currency display for mentors on signup page and homepage
Feb 19	Fixed admin panel currency formatting and email variable substitution
Feb 20	Fixed email funnel bugs and removed debug logs
Feb 20	Fixed student onboarding production environment issue

> **Week Summary:** *The sixth week was entirely devoted to designing and building a comprehensive email funnel system from the ground up.*

Week 7 (February 21 – February 27, 2026) — **Student Onboarding & Video Interview Reliability**

The seventh week delivered the complete student onboarding system and substantially improved the reliability of the video interview module. Redis was adopted for state management within the interview agent, eliminating the statelessness issues that had previously caused inconsistent interview sessions. The student onboarding flow was built end-to-end, incorporating additional profile fields such as date of birth, target year, budget with currency support, and standardized test scores. The student dashboard was updated to reflect onboarding progress through dedicated progress boards. Administrative capabilities were extended to allow administrators to view interview videos directly from the admin panel. Superadmin features were added to facilitate broader platform management, and a university agent architecture plan was formally drafted for future implementation.

Date	Task
Feb 21	Made video interview agent more reliable using Redis for state management
Feb 21	Fixed video interview flow bugs
Feb 21	Optimised email tracking token generation
Feb 21	Fixed sitemap access issue and added blocked user feature
Feb 22	Protected student onboarding form with authentication guard
Feb 22	Fixed student sign-in process
Feb 23	Built complete student onboarding flow with extra profile fields
Feb 23	Updated student dashboard to reflect onboarding state
Feb 23	Added date of birth field to settings page
Feb 23	Added admin panel view for interview videos
Feb 24	Fixed footer year display
Feb 25	Added target year field to student onboarding form
Feb 25	Added currency support for student budget field
Feb 26	Added standardised test scores field to student onboarding
Feb 26	Added onboarding progress boards to student dashboard
Feb 27	Added new superadmin features for platform management
Feb 27	Fixed student sign-out path in onboarding flow
Feb 27	Drafted university agent architecture plan

> **Week Summary:** *The seventh week delivered the complete student onboarding system and substantially improved the reliability of the video interview module.*

Week 8 (February 28 – March 6, 2026) — **University Profile System — Backend & Database**

The eighth week was dedicated to building the university profile system from the database layer up through to the frontend integration. The university profiles schema was designed and implemented with consideration for data integrity, including appropriate column size adjustments and a verification status field to support administrative review workflows. Backend APIs were developed and connected to the frontend, enabling the display of university profiles on both the homepage listing and dedicated profile pages. The admin panel was updated to include a direct link to university profiles for ease of administrative management. Additionally, critical bugs in the virtual interview module were resolved, including a stuck question issue and an interview progress refresh problem that had prevented users from resuming sessions seamlessly.

Date	Task
Feb 28	Updated platform architecture documentation
Mar 3	Added verification status field for universities
Mar 3	Updated university profiles schema and admin panel integration
Mar 3	Fixed build errors and updated database plan for university profiles
Mar 3	Fixed database column size issues for university data
Mar 4	Integrated university profile frontend with backend API
Mar 4	Increased JSON field size limit for universities data
Mar 4	Applied deployment fixes for university feature
Mar 5	Added university listing page to the homepage
Mar 5	Added university profile link in admin panel
Mar 5	Increased legal accreditation body VARCHAR limit in database
Mar 5	Fixed mentor setup and passport verification step
Mar 5	Added university name context to second interview question
Mar 5	Added interview progress continuity on page refresh
Mar 6	Fixed stuck question bug in virtual interview
Mar 6	Fixed interview progress refresh issue

> **Week Summary:** *The eighth week was dedicated to building the university profile system from the database layer up through to the frontend integration.*

Week 9 (March 7 – March 13, 2026) — **University Profile Frontend & Mentor Onboarding Fix**

The ninth week was focused on refining the university profile frontend and addressing outstanding defects in the mentor onboarding workflow. The university profile pages underwent UI and layout improvements to enhance usability and visual consistency across the platform. A TypeScript type error within the university profile feature was identified and corrected, ensuring type safety throughout the codebase. A significant bug in the mentor onboarding flow, specifically related to academic experience submission, was diagnosed and resolved, allowing mentors to complete their profiles without interruption. Project documentation, including the README and CONTRIBUTING guides, was updated to reflect the current state of the codebase and onboarding process for new developers.

Date	Task
Mar 8	Refined university profile pages UI and layout
Mar 9	Fixed TypeScript error in university profile feature
Mar 10	Fixed mentor onboarding academic experience submission bug
Mar 13	Added additional frontend files for university profile
Mar 13	Updated README and CONTRIBUTING documentation

> **Week Summary:** *The ninth week was focused on refining the university profile frontend and addressing outstanding defects in the mentor onboarding workflow.*

Week 10 (March 14 – March 20, 2026) — **University Profile Polish & Code Cleanup**

The tenth week concluded the university profile feature with final frontend refinements and a thorough code quality pass across the project. All remaining frontend changes for the university profile pages were completed and verified for production readiness. A gap report analysis was added to the university profile page, providing prospective students with structured insights into their application readiness relative to a given institution's requirements. A systematic removal of all console.log statements from frontend files was undertaken to eliminate debugging artefacts from the production build. General code cleanup and production readiness improvements were applied across the frontend codebase, reducing technical debt and improving maintainability.

Date	Task
Mar 14	Completed university page frontend changes
Mar 14	Added gap report analysis for university profile page
Mar 15	Removed all console.log statements from frontend files
Mar 15	Code cleanup and production readiness improvements

> **Week Summary:** *The tenth week concluded the university profile feature with final frontend refinements and a thorough code quality pass across the project.*

Week 11 (March 21 – March 27, 2026) — **Testing, Bug Fixes & Stability**

The eleventh week was entirely dedicated to quality assurance, with comprehensive end-to-end testing conducted across all major platform features. The mentor onboarding flow, student onboarding process, email funnel delivery, and university profile pages were each tested against a variety of user scenarios and edge cases to ensure functional correctness. Minor user interface inconsistencies identified across the student and mentor dashboards were corrected to maintain a cohesive user experience. University profile pages were validated across both mobile and desktop viewports to confirm responsive design compliance. All identified issues were formally documented with corresponding fix tickets created for subsequent resolution, establishing a structured backlog for upcoming sprints.

Date	Task
Mar 21	Reviewed and tested all major features end-to-end
Mar 22	Fixed edge cases in mentor onboarding flow
Mar 23	Tested student onboarding across different user scenarios
Mar 24	Verified email funnel delivery and tracking accuracy
Mar 25	Fixed minor UI inconsistencies across student and mentor dashboards
Mar 26	Tested university profile pages on mobile and desktop
Mar 27	Documented known issues and created fix tickets

> **Week Summary:** *The eleventh week was entirely dedicated to quality assurance, with comprehensive end-to-end testing conducted across all major platform features.*

Week 12 (March 28 – April 3, 2026) — **Performance Optimization & Refactoring**

The twelfth week was oriented towards performance optimisation and architectural improvement across the platform's backend and frontend systems. API response times were audited to identify slow database queries, with Prisma queries subsequently refactored to use precise field selection via select and include clauses, reducing unnecessary data transfer. The mentor search service was refactored for improved performance, and unused dependencies were removed from various packages to reduce build size. React Query caching strategies within the student dashboard were improved to minimise redundant network requests and unnecessary component re-renders. The email service was refactored with enhanced error handling to improve resilience and debuggability. The week concluded with a comprehensive code review and pull request cleanup session.

Date	Task
Mar 28	Audited API response times and identified slow queries
Mar 29	Optimised Prisma queries with proper select/include fields
Mar 30	Refactored mentor search service for better performance
Mar 31	Cleaned up unused dependencies across packages
Apr 1	Improved React Query caching strategies in student dashboard
Apr 2	Refactored email service for better error handling
Apr 3	Code review and PR cleanup

> **Week Summary:** *The twelfth week was oriented towards performance optimisation and architectural improvement across the platform's backend and frontend systems.*

Week 13 (April 4 – April 10, 2026) — **Admin Panel Enhancements**

The thirteenth week was focused on enhancing the administrative panel with advanced analytics, workflow improvements, and data management capabilities. Analytics charts powered by the Recharts library were integrated into the admin dashboard, providing visual summaries of platform activity. The mentor verification workflow was streamlined to accelerate the administrative review and approval process. Bulk action support was introduced for mentor approval and rejection operations, significantly reducing the time required to process large volumes of pending applications. Email campaign analytics were enriched with open rate and click rate metrics, enabling data-driven assessment of campaign effectiveness. An export functionality was implemented for both student and mentor data, supporting reporting and compliance requirements. Pagination issues in the admin panel were resolved and all new features were thoroughly tested.

Date	Task
Apr 4	Added analytics charts to admin dashboard using Recharts
Apr 5	Improved mentor verification workflow in admin panel
Apr 6	Added bulk action support for mentor approval/rejection
Apr 7	Enhanced email campaign analytics — open rate, click rate display
Apr 8	Added export functionality for student and mentor data
Apr 9	Fixed admin panel pagination issues
Apr 10	Tested and verified all admin panel features

> **Week Summary:** *The thirteenth week was focused on enhancing the administrative panel with advanced analytics, workflow improvements, and data management capabilities.*

Week 14 (April 11 – April 17, 2026) — **Chat System & Real-time Features**

The fourteenth week concentrated on improving the reliability and user experience of the platform's real-time chat system. The existing WebSocket chat implementation was reviewed in detail to identify stability vulnerabilities, and message delivery reliability was subsequently improved to prevent message loss during high-latency conditions. An unread message count badge was added to the chat interface, providing users with clear visibility of pending communications. The chat session creation flow between students and mentors was refined to reduce friction during initial connection. Chat history loading on session resume was corrected to ensure continuity of conversation context. A typing indicator was introduced to provide real-time feedback during active conversations, and the chat system was tested under concurrent user load to verify stability.

Date	Task
Apr 11	Reviewed WebSocket chat implementation for stability
Apr 12	Fixed message delivery reliability in chat sessions
Apr 13	Added unread message count badge to chat UI
Apr 14	Improved chat session creation flow between student and mentor
Apr 15	Fixed chat history loading on session resume
Apr 16	Added typing indicator to real-time chat
Apr 17	Tested chat system under concurrent user load

> **Week Summary:** *The fourteenth week concentrated on improving the reliability and user experience of the platform's real-time chat system.*

Week 15 (April 18 – April 24, 2026) — **Referral System & Payment Integration**

The fifteenth week addressed the referral and payment systems, ensuring both functional correctness and data integrity. The referral code generation and tracking logic was reviewed, and a bug in the earnings calculation for multi-level referral scenarios was identified and corrected. The Razorpay payment flow was tested end-to-end, encompassing payment initiation, webhook reception, and database record updates. Edge cases in webhook signature verification were resolved to prevent potential spoofing vulnerabilities. A referral withdrawal request flow was developed for mentors, enabling them to formally request disbursement of accrued referral earnings. The referral dashboard was enhanced with a detailed earnings breakdown view, and database consistency between payment records and referral data was verified.

Date	Task
Apr 18	Reviewed referral code generation and tracking logic
Apr 19	Fixed referral earnings calculation for multi-level referrals
Apr 20	Tested Razorpay payment flow end-to-end
Apr 21	Fixed webhook signature verification edge cases
Apr 22	Added referral withdrawal request flow for mentors
Apr 23	Improved referral dashboard UI with earnings breakdown
Apr 24	Verified payment and referral data consistency in database

> **Week Summary:** *The fifteenth week addressed the referral and payment systems, ensuring both functional correctness and data integrity.*

Week 16 (April 25 – May 1, 2026) — **Final Review, Documentation & Deployment**

The sixteenth and final week of the internship was devoted to production readiness activities, ensuring the platform was fully stable and comprehensively documented prior to handover. A final round of end-to-end testing was conducted across all platform features to identify and resolve any remaining production defects. API documentation and environment variable references were updated to reflect the current system configuration, providing a reliable reference for future developers. The production build configuration was optimised across all Turborepo packages to reduce build times and artefact sizes. PM2 process management and deployment scripts were verified to ensure consistent and reliable server operation. A comprehensive final code review was conducted across all packages, followed by the preparation of internship completion documentation and detailed handover notes.

Date	Task
Apr 25	Conducted final end-to-end testing of all platform features
Apr 26	Fixed remaining production bugs identified during testing
Apr 27	Updated API documentation and environment variable reference
Apr 28	Optimised production build configuration for all apps
Apr 29	Verified PM2 process management and deployment scripts
Apr 30	Final code review and cleanup across all packages
May 1	Prepared internship completion report and handover notes

> **Week Summary:** *The sixteenth and final week of the internship was devoted to production readiness activities, ensuring the platform was fully stable and comprehensively documented prior to handover.*